Winston Li

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Education

UC Irvine • B.S. in Computer Science w/ Honors, 3.9 / 4.0 GPA — Sep 2024 – Jun 2026

- Graduated in 2 years • ICS Honors Program • UROP Fellow
- Data Structures • Algorithms • Software Eng. • Operating Systems • Compilers
- Cryptography • Formal Lang. & Automata • Quantum Computation • Distributed Systems

Projects

Languages & compilers • *Rust, C99, WebAssembly* — Feb 2023 – Present

- **tatic**: Engineering a **JIT** caching **WebAssembly** by content hash; installs code only if it matches an interpreter and has a **machine-checked proof** of the compiler model.
  - Fuzzing it against the interpreter (~12k trials, no mismatches); caught a checker overflow and a leak.
- **cereal**: Writing a C99 front end + **language server**; matching gcc-13 on 3,908/3,910 gcc.dg tests.
- **renno**: Designing a CEK-machine language with **algebraic effects** and gradual typing; 400+ tests.
- **Lambast** (2025, Honors): Wrote optimizing λ-calculus compiler (**closure conversion**); UROP talk.
- **CS 142B project**: Built an SSA compiler (Braun et al.'s **marker algorithm**); CSE, phi simplification.
- **renno** (2023, prototype): Developed a Rust-to-JS compiler with name resolution and a type checker.

scratchtape • *Rust, wgpu, cubecl*

- Developing an ML engine from scratch (autodiff, layers, GPU); gradient-checking each primitive.
- Measuring **DX12** at 1.6–5.8× lower overhead than Vulkan on small matmuls; up to 900 GFLOP/s.

manifold • *Rust, wgpu, CEF, Servo*

- Building an 85k-line browser: an engine process per tab (Blink, Servo, WebKit) over IPC; hot-swap.
- Adding crash recovery, journaled sessions, and a force-directed tab graph on **wgpu**.

plumb • *Rust, Windows, NTFS*

- Writing a disk-usage analyzer that reads the **NTFS MFT** to scan a 4.3M-node volume in under 10 s.
- Integrating live updates from the NTFS change journal and a partial tree while scanning; 200+ tests.

arrow-game • *TypeScript, Svelte, Tauri*

- Crafting a strict-TS roguelite; 400+ tests; balancing it with an evolved bot that lasts 3× longer.

Skills and interests

Proficient in Rust, C / C++, JS / TS, Python, Svelte, Tailwind.

Familiar with Git, HTML / CSS, Nix, LaTeX, Typst, Claude Code, Unix systems & shells.

Acquainted with most mainstream languages, e.g., C#, Go, Haskell, Java, Lua, Ruby, SQL.

Experience

Learning Assistant • UC Irvine — Mar 2025 – Jun 2026

- Delivered supplementary pedagogy for **Python** and upper-div **Algorithms,** e.g., Master Theorem.
- Spearheaded study sessions to identify and guide students through gaps in understanding.

Intern • LGBTQIA+ Identity Commission @ UC Irvine — Oct 2024 – Jun 2025

- Led a team of 3 to organize and host themed Gayme Show™.

Programmer • Video Game Development Club @ UC Irvine — Sep 2024 – May 2025

- Worked with team of 13 to create typing game in **C#** and **Unity**; led dev of the main UI window.